

SYLLABUS OF EMPIRICAL INDUSTRIAL ORGANIZATION I (GRADUATE)

Fall 2026

Instructor:	Faqiang Li	Time:	Tues. 3:10-6:00
Office Hour:	By appointment	Place:	2 Jiao 511
Email:	faqiangli.econ@nsd.pku.edu.com	TA:	Wende Zhang wdzhang2019@nsd.pku.edu

Description This is a semester-long graduate-level course on empirical industrial organization. The course aims to provide a comprehensive understanding of firms' production, consumers' decisions, market structures, and the strategic interactions among firms and consumers. It addresses pressing policy concerns such as industrial policy, innovation, consumer welfare, antitrust, and regulation through rigorous quantitative modeling. Additionally, the analytical framework and knowledge are practically relevant in real-world business scenarios, including the analysis of consumer behavioral patterns, startup strategy, pricing strategies, and industrial competition. Therefore, the course is expected to broaden students' knowledge of theoretical and empirical tools in modern industrial organization research, develop their ability to analyze an industry in a real-world business setting, and, most importantly, cultivate their ability to conduct independent research.

The class will be divided into four modules:

- i) An overview of the field, core questions, and discoveries
- ii) Modern methodology and application of production function estimation
- iii) Characteristic-based demand estimation and its application in typical scenarios such as welfare analysis, merger analysis, and the analysis of vertical structures
- iv) Models in industry dynamics, dynamic games, and their estimation

The class requires a solid background in first-year Ph.D. economics, including topics such as Nash equilibrium, dynamic models of investment, Cournot and Bertrand models, endogeneity, and GMM estimation. Students must be familiar with a statistical package (e.g., Stata or R) and a matrix programming language (e.g., MATLAB, Julia, R, Python). It is highly recommended to attend course like computational economics before or alongside this course. Datasets for production function estimation and demand estimation will be provided for homework.

Gen-AI Protocol: Integrate gen-AI to your research pipeline is acceptable and encouraged. But students need to be fully responsible for all your submitted work. And do note that, contents produced by gen-AI might be a poor substitute for your own thinking.

AI in Econ-Research Workshop I will hold a 2-3 hour workshop on my own experience in using AI tools in economic research, around mid September, focusing on context engineering, research management, and some real user cases. Time TBA. Voluntary participation. Spread the words!

Office Hours: There is no fixed time. You are encouraged to make an appointment by email. And you should take it very seriously for meeting with faculty (not just me). See the appendix for John Asker's suggestions on preparing for a meeting with faculty.

Presentation: We will discuss this after the first class and make a decision before the second one.

Research Proposal: See the guide provided separately. The tentative deadline will be Dec. 31. Each student is required to meet me and discuss your preliminary proposal more than once during the mid-term

weeks. Eventually, each student should submit their own proposal, although students are encouraged to share data sources. Two people working on a same project and submitting the same proposal is not allowed.

Textbooks: There is no textbook for this course, but the following are good references:

- Empirical: Akerberg, Lanier Benkard, Berry, and Pakes (2007)
- Theory: Tirole's *Theory of Industrial Organization*

General Guidelines for Students

- Read this: do you REALLY want to have a Ph.D. degree? Try this: https://www.sohu.com/a/513039020_121073572. If yes, then this is useful: <https://zhuanlan.zhihu.com/p/454686562> or <http://li.dyson.cornell.edu/phdRes.php>.
- Sign up to receive weekly updates of the latest working papers from the National Bureau of Economic Research (<http://www.nber.org/prefs/Notify>). Being a researcher requires knowing the economist community and literature. Read, at minimum, the abstracts of all papers in your area of interest each week! Also, pick your fav journal, e.g., AER (<https://www.aeaweb.org/notify/>), and do the same thing.
- Join the weekly applied seminars of your interest from all PKU departments. Pay attention to what the presenters choose to emphasize. Pay attention to the critiques and feedback from the audience. What do the presenters expect the audience to ask? Understanding how to present a paper and anticipating the issues that others are likely to care about is an important part of your training. Also, ask questions in the seminar!
- Regarding AI in research, keep these in minds
 1. The awareness to use these tools
 2. The wisdom to curb your use. See <https://time.com/7295195/ai-chatgpt-google-learning-school/>. And be concerned about potentially reducing chances to learn from your professors https://erikhurst.com/wp-content/uploads/2026/04/automation_learning_nber.pdf
 3. The ability to CLARIFY your needs (in Chinese or English) is crucial now for researchers.
 4. Generate the BIG PICTURE of your ideas and break them down to IMPLEMENTABLE sub-steps.
 5. Yes, you will have more projects than what you can handle. Organize your energy and research like a project manager.
- Learn to manage your research
 1. Ideas generation
 - (a) Source of IO ideas?
 - Your curiosity
 - News about markets and industries (e.g., <https://www.cnbc.com/2025/01/24/how-chinas-new.html>)
 - Pay attention to antitrust case, e.g., check this paper out, Xia (2024)
 - Self generation and cultivation, e.g., keep writing notes on a notebook app like Obsidian
 2. Join seminars and take notes. For PKU students, sign up for this PKU-Tsinghua econ seminar list <https://docs.qq.com/sheet/DUVZZektqV2ViS255?tab=BB08J2>.*
 3. Daily reading and literature management

*The link may be expire. Ask your friends how to get the seminar feed.

- (a) Email, journal email subscription, as mentioned above
- (b) Zotero, Mendeley, and maybe more advanced AI tools
- 4. Management across projects
 - (a) Workspace (e.g., Dropbox)
 - (b) Project management tools
- 5. Management within a project
 - (a) Doing research is like managing a team. There will be many non-academic problems you can solve to boost your productivity, e.g., thinking about version controls. For a systematic reference, check [Code and data for the social sciences: A practitioner's guide](#). Or alternatively, https://benjaminvatter.com/uploads/coding_prep.pdf
 - (b) GitHub, or any other tool you use to keep track of versions of your code
 - (c) Overleaf
 - (d) Calendar
 - (e) Using a high-performance computer cluster to conduct parallel computing
 - (f) These days, integrated workflow with AI tools are highly relevant

General Reference for Students in IO

- Structural econometrics: what and why
 - Chapter 64: Structural Econometric Modeling: Rationales and Examples from Industrial Organization, sections 1, 2, 3, 4
 - Haile (2019) slides: Structural vs. Reduced Form: Language and Models in Empirical Economics
 - Heckman and Vytlacil (2005): "Structural equations, treatment effects, and econometric policy evaluation." *Econometrica* 73, no. 3 (2005): 669-738.
 - Galiani and Pantano (2021)
- Sources of trendy IO papers (or any field)
 - Authors' webpages
 - Field conferences: IIOC, EARIE, APIOS sessions, or Chinese IO workshops (SHUFE IO <https://cob.sufe.edu.cn/websites/Summeractivities/2025/2024-en.html>). By the way, you can find (almost) all the conferences here: <https://editorialexpress.com/cgi-bin/conference/cm.cgi> — aim some conference to boost your productivity!
 - Conferences with highest popularity: NBER IO (summer, fall, winter) meetings
- How to find trendy topics in IO (or any field)
 - Find out who hosts NBER IO meetings and follow their work
 - Find out which papers are listed in the NBER spring, summer, fall programs
 - Check out the rising star session of IIOC, EARIE
 - Explore the latest IO JMPs: shoshanavasserman.com

Prize: Yes, competition and commitment make better research! If you have an innovative and practical (i.e., implementable) proposal, then Faqiang promises to give you a prize to waive your registration fee for presenting in APIOS/Earie/IIOC. The prize is based on the quality of your proposals.

Grading: Grading for the course will be based on four components:

1. *Class Participation and Reading 10%*: I will evaluate in-class participation of each student. Attendance is necessary but not sufficient for participation. Reading is very important. Students are expected to read the starred (*) and dagger (†) papers on the reading list each week. The starred (*) papers will be covered in class lectures, and dagger (†) papers will be presented by students. Displaying knowledge of other papers on the reading list or related papers not on the reading list will improve your participation and reading score.
2. *Presentations 30%*: The papers marked with a dagger (†) will be presented by students. Presentations will be 30-35 minutes and the rest of the time in that class is Q&A and discussion. The time limit is strictly enforced. Multiple students will prepare the presentations and they are the candidates for the presentation. Only one presenter will be determined on the presentation day by a random draw. The presentation will be evaluated based on the slides, delivery, and time control. Other non-presenting candidates are responsible for providing comments and thoughts during and after the presentations. Their slides and participation will be evaluated. On average, students will need to prepare for two to three presentations during the whole semester. Details are on a separate handout.
3. *Term Proposal 30%*: Students will complete a proposal for an empirical project no longer than 10 A4 pages. Students should not turn an existing proposal with mature implementation into a new proposal. All proposals are welcome, but IO proposals are encouraged. Details will be provided on the first day of class. The final proposal will be due at noon on December 25. In the middle of the semester after sufficient entry into the demand module, every student will present their proposal for 7 minutes, and the rest of the class will provide feedback. We will find a separate class for this based on the final size of the class.
4. *Problem Sets and Referee Report 30%*: Two coding problem sets and one or two other small quizzes will be given (production function estimation and demand estimation). The problem sets are assigned because, to really understand a topic, you need to be able to replicate some of the work. Students can work on problem sets in groups of up to 3 members. Each student must write their own code and solution. The referee report will be on the papers from this semester's NSE speakers in the field of IO (see below). You can choose either one. You need to submit the referee report BEFORE the seminar starts — the point is to engage yourself and enable you to ask incisive questions during the seminars. After negotiating with me, you may also choose seminars on an IO topic from other departments of PKU, Tsinghua, or Renmin (there are a lot!). In such cases, you also need to submit the report before the seminar.

Important Dates: (Tentative)

Preliminary pb start	Sept 9
Sign-up for presentations	Sept 12
Panle Jia (Wisconsin) seminar	Sept 18
Preliminary pb due	Sept 23
Coding Pb1 start	Sept 23
Coding Pb1 due	Oct 14
Mid-term proposal meeting with Faqiang ...	Oct 12–19
Han Ng (Academia Sinica) seminar	Dec 15
Coding Pb2 start	Nov 4
Coding Pb2 due	Nov 25
Proposal Due	Dec 31

Tentative Course Outline:

1. Introduction to Empirical IO

- Introductory Lecture: Course Overview, Rise of Global Market Power, and the Field of IO

- *Berry, Gaynor, and Morton (2019)
- De Loecker and Eeckhout (2021)
- Döpper, MacKay, Miller, and Stiebale (2021)
- Shapiro (2019)
- Syverson (2019)
- Miller (2024)
- IO “retrospective” event: <https://econincubator.uchicago.edu/events/industrial-organizational-economics-retrospective>

2. Production Function: Model and Empirical Implementation

• Production Function I: Data, Findings, Models

- *What is a typical “production dataset”: Census of Manufacturing Questionnaire (U.S. and China versions)
- *Asker, Collard-Wexler, and De Loecker (2014) (based on Bloom (2009))
- *Chapter 1, 2, 4 of De Loecker and Syverson (2021)
- Dunne, Roberts, and Samuelson (1988) and Dunne, Roberts, and Samuelson (1989)
- China context: Brandt, Van Biesebroeck, Wang, and Zhang (2017)
- Distinguish macro and IO “TFP”: Hsieh and Klenow (2009)

• Production Function II: Problems in Estimation and Modern Solution

- * Olley and Pakes (1996)
- *Akerberg, Caves, and Frazer (2015)
- Levinsohn and Petrin (2003)
- Pavcnik (2003)

• Production Function III: More on Methodology and Typical Applications (Productivity, Markup, and Revenue-based data)

- * Loecker (2011)
- * Loecker and Warzynski (2012)
- Gandhi, Navarro, and Rivers (2020)
- De Loecker, Eeckhout, and Unger (2020)
- Collard-Wexler and De Loecker (2015)
- Grieco, Li, and Zhang (2022)
- Demirer (2022)

• Production IV: More Applications (Two presentations)

- *De Loecker, Goldberg, Khandelwal, and Pavcnik (2016)
- Allcott, Collard-Wexler, and O’Connell (2016)
- Collard-Wexler and De Loecker (2015)
- ~~– †Aw, Roberts, and Xu (2011)~~
- †Rubens (2023)
- Rubens (2025)
- Demirer (2020)

3. Demand Estimation: Model and Empirical Implementation

• Demand Estimation I: Demand Estimation via a History of “New Empirical IO”

- *A history of New Empirical IO: Bresnahan (1982)
- *Deaton and Muellbauer (1980)

- *Hausman, Leonard, and Zona (1994),
- What is a typical “demand dataset”: Texas car market example
- Nash-Bertrand pricing model and how to analyze an industry
- Pinkse, Slade, and Brett (2002)
- Corts (1999)
- **Demand Estimation II: BLP Methodology**
 - *Berry (1994)
 - *Berry, Levinsohn, and Pakes (1995)
 - Identification: Gandhi and Houde (2019), Berry and Haile (2024a)
 - Computation and guide: Nevo (2000a), Dubé, Fox, and Su (2012), Conlon and Gortmaker (2020)
 - Revisit the product space methods: Conlon and Mortimer 2024 (draft not publicly available)
- **Demand Estimation III: BLP Application: Welfare Analysis, Conduct Testing, and Dealing with Microdata**
 - *Petrin (2002)
 - *Berto Villas-Boas (2007)
 - *Nevo (2001)
 - Berry, Levinsohn, and Pakes (1999)
 - Ito and Zhang (2020)
 - The following topics are important but they will not be included in this class. Learn yourself.
 - * **Incorporate Consumer Level Choice Data**
 - Murry (2017)
 - Grieco, Murry, and Yurukoglu (2021)
 - Berry, Levinsohn, and Pakes (2004)
 - Hastings, Kane, and Staiger (2005)
 - * **Demand Estimation: Recent Progress in Demand Estimation**
 - Gandhi and Houde (2026)
 - Duarte, Magnolfi, Sølvsten, and Sullivan (2024)
 - Berry and Haile (2024b)
 - Grieco, Murry, Pinkse, and Sagl (2025)
 - Miravete, Seim, and Thurk (2026)
 - Goldszmidt, Henry de Frahan, List, Mogstad, Muir, and Torgovitsky (2026)
- **Demand Estimation IV: BLP Application: Merger and Anti-trust (One presentation)**
 - This year, we are thrilled to have Zhou Zhong to talk about Chinese anti-trust practice!
 - * Nevo (2000b)
 - †Miller and Weinberg (2017)
 - Backus, Conlon, and Sinkinson (2021)
 - Illanes, Bhattacharya, Stillerman, Kreps, and Salas (2024)
 - What are the common statistics of assessing a merger? Read U.S. DOJ: Horizontal Merger Guidelines (08/19/2010), and also a recent comment https://economics.mit.edu/sites/default/files/2022-09/What%20Next%20for%20the%20Horizontal%20Merger%20Guidelines_.pdf
 - Gowrisankaran, Nevo, and Town (2015)
 - Ciliberto and Williams (2014)
- **Demand Estimation V: BLP Application: Bargaining and vertical relationship (One Presentation)**

- Crawford, Lee, Whinston, and Yurukoglu (2018)
- *Crawford and Yurukoglu (2012)
- Lee (2013)
- Grennan (2013)
- Ho and Lee (2017)
- †Xia (2024)
- ~~– †Demirer and Rubens (2025)~~
- Ho and Lee (2019)

4. Demand Estimation IV: BLP Application: Endogenous Characteristics (Two Presentation)

- * Fan (2013)
- † Gentzkow, Shapiro, and Sinkinson (2014)
- ~~• † Wollmann (2018)~~
- Eizenberg (2014)
- Hoberg and Phillips (2022)

5. Dynamic Model: Model, Estimation, and Simulation

- **Dynamic Model I: Rust and Hotz&Miller Conditional Choice Probability Estimators**
 - *Rust (1987)
 - *Hotz and Miller (1993)
 - Magnac and Thesmar (2002)
- **Dynamic Model II: Extension in Estimation Methodology**
 - *Houde, Newberry, and Seim (2023)
 - *Arcidiacono and Ellickson (2011)
 - *Holmes (2011)
 - Peter Arcidiacono and Miller (2011)
 - Arcidiacono, Bayer, Blevins, and Ellickson (2016)
- **Dynamic Model III: Estimation of Entry Model(One presentation)**
 - *Seim (2006)
 - ~~– †Singleton (2019)~~
 - Bresnahan and Reiss (1990)
 - Berry (1992)
 - *Ciliberto and Tamer (2009)
- **Dynamic Model IV: Markov Perfect Nash Equilibrium and Estimation**
 - *Ryan (2012)
 - *Sweeting (2013)
 - Bajari, Benkard, and Levin (2007)
 - Ifrach and Weintraub (2017)
 - Pakes, Ostrovsky, and Berry (2007)
 - Bajari, Chernozhukov, Hong, and Nekipelov (2015)
 - Dunne, Klimek, Roberts, and Xu (2013)
- **Dynamic Model V: Industrial Policy and Other Applications (Two presentations)**
 - *Barwick, Kalouptsi, and Zahur (2019)
 - Garg and Saxena (2023)

- Buchholz (2022)
- Jeon (2022)
- Bodere (2022)
- Caoui (2022)
- †Sabal (2025)

References

- Akerberg, Daniel et al. (2007). “Chapter 63 Econometric Tools for Analyzing Market Outcomes”. en. In: *Handbook of Econometrics*. Vol. 6, pp. 4171–4276. ISBN: 978-0-444-50631-3.
- Akerberg, Daniel A., Caves, Kevin, and Frazer, Garth. “Identification Properties of Recent Production Function Estimators”. *Econometrica*, 2015, 83, 6, pp. 2411–2451.
- Allcott, Hunt, Collard-Wexler, Allan, and O’Connell, Stephen D. “How Do Electricity Shortages Affect Industry? Evidence from India”. *American Economic Review*, 2016, 106, 3, pp. 587–624.
- Arcidiacono, Peter and Ellickson, Paul B. “Practical Methods for Estimation of Dynamic Discrete Choice Models”. *Annual Review of Economics*, 2011, 3, 1, pp. 363–394.
- Arcidiacono, Peter et al. “Estimation of Dynamic Discrete Choice Models in Continuous Time with an Application to Retail Competition”. *The Review of Economic Studies*, 2016, 83, 3, pp. 889–931.
- Asker, John, Collard-Wexler, Allan, and De Loecker, Jan. “Dynamic Inputs and Resource (Mis)Allocation”. *Journal of Political Economy*, 2014, 122, 5, pp. 1013–1063.
- Aw, Bee Yan, Roberts, Mark J, and Xu, Daniel Yi. “R&D investment, exporting, and productivity dynamics”. *American Economic Review*, 2011, 101, 4, pp. 1312–1344.
- Backus, Matthew, Conlon, Christopher, and Sinkinson, Michael (Jan. 2021). *Common Ownership and Competition in the Ready-to-Eat Cereal Industry*. Working Paper.
- Bajari, Patrick, Benkard, C. Lanier, and Levin, Jonathan. “Estimating Dynamic Models of Imperfect Competition”. *Econometrica*, 2007, 75, 5, pp. 1331–1370.
- Bajari, Patrick et al. (Apr. 2015). *Identification and Efficient Semiparametric Estimation of a Dynamic Discrete Game*. en. Tech. rep. w21125. National Bureau of Economic Research, w21125.
- Barwick, Panle Jia, Kalouptsi, Myrto, and Zahur, Nahim Bin (2019). *China’s industrial policy: An empirical evaluation*. Tech. rep. National Bureau of Economic Research.
- Berry. “Estimating discrete-choice models of product differentiation”. *The RAND Journal of Economics*, 1994, pp. 242–262.
- Berry, Steven, Gaynor, Martin, and Morton, Fiona Scott. “Do Increasing Markups Matter? Lessons from Empirical Industrial Organization *”. *Journal of Economic Perspectives*, 2019, 33(3), pp. 44–68.
- Berry, Steven, Levinsohn, James, and Pakes, Ariel. “Automobile Prices in Market Equilibrium”. *Econometrica*, 1995, 63, 4, p. 841.
- Berry, Steven, Levinsohn, James, and Pakes, Ariel. “Voluntary Export Restraints on Automobiles: Evaluating a Trade Policy”. *American Economic Review*, 1999, 89, 3, pp. 400–430.
- Berry, Steven, Levinsohn, James, and Pakes, Ariel. “Differentiated Products Demand Systems from a Combination of Micro and Macro Data: The New Car Market”. *Journal of Political Economy*, 2004, 112, 1, pp. 68–105.
- Berry, Steven T. “Estimation of a Model of Entry in the Airline Industry”. *Econometrica*, 1992, 60, 4, p. 889.
- Berry, Steven T. and Haile, Philip A. “Nonparametric Identification of Differentiated Products Demand Using Micro Data”. *Econometrica*, 2024, 92, 4, pp. 1135–1162.
- Berry, Steven T. and Haile, Philip A. “Nonparametric Identification of Differentiated Products Demand Using Micro Data”. *Econometrica*, 2024, 92, 4, pp. 1135–1162.
- Berto Villas-Boas, Sofia. “Vertical Relationships between Manufacturers and Retailers: Inference with Limited Data”. *The Review of Economic Studies*, 2007, 74, 2, pp. 625–652.
- Bloom. “The Impact of Uncertainty Shocks”. *Econometrica*, 2009, 77, 3, pp. 623–685.
- Bodere, Pierre. “Dynamic Spatial Competition in Early Education: an Equilibrium Analysis of the Preschool Market in Pennsylvania”, 2022.

- Brandt, Loren et al. "WTO Accession and Performance of Chinese Manufacturing Firms". *American Economic Review*, 2017, 107, 9, pp. 2784–2820.
- Bresnahan, Timothy F. "The oligopoly solution concept is identified". *Economics Letters*, 1982, 10, 1-2, pp. 87–92.
- Bresnahan, Timothy F. and Reiss, Peter C. "Entry in Monopoly Markets". *The Review of Economic Studies*, 1990, 57, 4, p. 531.
- Buchholz, Nicholas. "Spatial Equilibrium, Search Frictions, and Dynamic Efficiency in the Taxi Industry". *The Review of Economic Studies*, 2022, 89, 2, pp. 556–591.
- Caoui, El Hadi. "Estimating the Costs of Standardization: Evidence from the Movie Industry". *Review of Economic Studies*, 2022.
- Ciliberto, Federico and Tamer, Elie. "Market Structure and Multiple Equilibria in Airline Markets". *Econometrica*, 2009, 77, 6, pp. 1791–1828.
- Ciliberto, Federico and Williams, Jonathan W. "Does multimarket contact facilitate tacit collusion? Inference on conduct parameters in the airline industry". *The RAND Journal of Economics*, 2014, 45, 4, pp. 764–791.
- Collard-Wexler, Allan and De Loecker, Jan. "Reallocation and Technology: Evidence from the US Steel Industry". *American Economic Review*, 2015, 105, 1, pp. 131–171.
- Conlon, Christopher and Gortmaker, Jeff. "Best practices for differentiated products demand estimation with PyBLP". *The RAND Journal of Economics*, 2020, 51, 4, pp. 1108–1161.
- Corts, Kenneth S. "Conduct parameters and the measurement of market power". *Journal of Econometrics*, 1999, 88, 2, pp. 227–250.
- Crawford, Gregory S and Yurukoglu, Ali. "The Welfare Effects of Bundling in Multichannel Television Markets". *American Economic Review*, 2012, 102, 2, pp. 643–685.
- Crawford, Gregory S. et al. "The Welfare Effects of Vertical Integration in Multichannel Television Markets". *Econometrica*, 2018, 86, 3, pp. 891–954.
- De Loecker, Jan and Eeckhout, Jan (2021). *Global Market Power*.
- De Loecker, Jan, Eeckhout, Jan, and Unger, Gabriel. "The Rise of Market Power and the Macroeconomic Implications". *The Quarterly Journal of Economics*, 2020, 135, 2, pp. 561–644.
- De Loecker, Jan and Syverson, Chad (Jan. 2021). "An industrial organization perspective on productivity". In: *Handbook of Industrial Organization*. Ed. by Kate Ho, Ali Hortaçsu, and Alessandro Lizzeri. Vol. 4. Handbook of Industrial Organization, Volume 4, pp. 141–223.
- De Loecker, Jan et al. "Prices, Markups, and Trade Reform". *Econometrica*, 2016, 84, 2, pp. 445–510.
- Deaton, Angus and Muellbauer, John. "An Almost Ideal Demand System". *American Economic Review*, 1980, 70, 3, pp. 312–26.
- Demirer, Mert. "Production function estimation with factor-augmenting technology: An application to markups". *Job Market Paper*, 2020, 1, 2.2, p. 3.
- Demirer, Mert. "Production Function Estimation with Factor-Augmenting Technology: An Application to Markups", 2022.
- Demirer, Mert and Rubens, Michael. "Welfare Effects of Buyer and Seller Power", 2025, w33371.
- Döpper, Hendrik et al. "Rising Markups and the Role of Consumer Preferences". *SSRN Electronic Journal*, 2021.
- Duarte, Marco et al. "Testing Firm Conduct". *Quantitative Economics*, 2024, 15, 3, pp. 571–606.
- Dubé, Jean-Pierre, Fox, Jeremy T., and Su, Che-Lin. "Improving the Numerical Performance of Static and Dynamic Aggregate Discrete Choice Random Coefficients Demand Estimation". *Econometrica*, 2012, 80, 5, pp. 2231–2267.
- Dunne, Timothy, Roberts, Mark J., and Samuelson, Larry. "Patterns of Firm Entry and Exit in U.S. Manufacturing Industries". *The RAND Journal of Economics*, 1988, 19, 4, p. 495.
- Dunne, Timothy, Roberts, Mark J., and Samuelson, Larry. "The Growth and Failure of U. S. Manufacturing Plants*". *The Quarterly Journal of Economics*, 1989, 104, 4, pp. 671–698.
- Dunne, Timothy et al. "Entry, exit, and the determinants of market structure: Entry, exit, and the determinants of market structure". *The RAND Journal of Economics*, 2013, 44, 3, pp. 462–487.

- Eizenberg, Alon. “Upstream Innovation and Product Variety in the U.S. Home PC Market”. *The Review of Economic Studies*, 2014, 81, 3, pp. 1003–1045.
- Fan, Ying. “Ownership Consolidation and Product Characteristics: A Study of the US Daily Newspaper Market”. *American Economic Review*, 2013, 103, 5, pp. 1598–1628.
- Galiani, Sebastian and Pantano, Juan (2021). *Structural Models: Inception and Frontier*. NBER Working Paper w28698. National Bureau of Economic Research.
- Gandhi, Amit and Houde, Jean-François (Oct. 2019). *Measuring Substitution Patterns in Differentiated-Products Industries*. en. Tech. rep. w26375. National Bureau of Economic Research, w26375.
- (2026). “Measuring Substitution Patterns in Differentiated-Products Industries”.
- Gandhi, Amit, Navarro, Salvador, and Rivers, David A. “On the Identification of Gross Output Production Functions”. *Journal of Political Economy*, 2020, 128, 8, pp. 2973–3016.
- Garg, Shresth and Saxena, Sagar. “Dynamic Effects of Price Controls and Deregulation Policies: Evidence from the Indian Cement Industry”, 2023.
- Gentzkow, Matthew, Shapiro, Jesse M., and Sinkinson, Michael. “Competition and Ideological Diversity: Historical Evidence from US Newspapers”. *American Economic Review*, 2014, 104, 10, pp. 3073–3114.
- Goldszmidt, Ariel et al. (2026). “On the Identification and Interpretation of Price Elasticities in a Class of Discrete-Choice Demand Models”.
- Gowrisankaran, Gautam, Nevo, Aviv, and Town, Robert. “Mergers When Prices Are Negotiated: Evidence from the Hospital Industry”. *American Economic Review*, 2015, 105, 1, pp. 172–203.
- Grennan, Matthew. “Price Discrimination and Bargaining: Empirical Evidence from Medical Devices”. *American Economic Review*, 2013, 103, 1, pp. 145–177.
- Grieco, Paul L. E., Li, Shengyu, and Zhang, Hongsong. “Input Prices, Productivity, and Trade Dynamics: Long-Run Effects of Liberalization on Chinese Paint Manufacturers”. *The RAND Journal of Economics*, 2022, 53, 3, pp. 516–560.
- Grieco, Paul L. E., Murry, Charles, and Yurukoglu, Ali (July 2021). *The Evolution of Market Power in the US Auto Industry*. Working Paper.
- Grieco, Paul L. E. et al. (2025). *Optimal Estimation of Discrete Choice Demand Models with Consumer and Product Data*. NBER Working Paper 33397. National Bureau of Economic Research.
- Hastings, Justine, Kane, Thomas, and Staiger, Douglas (Nov. 2005). *Parental Preferences and School Competition: Evidence from a Public School Choice Program*. en. Tech. rep. w11805. National Bureau of Economic Research, w11805.
- Hausman, Jerry, Leonard, Gregory, and Zona, J. Douglas. “Competitive Analysis with Differentiated Products”. *Annales d’Économie et de Statistique*, 1994, 34, pp. 159–180.
- Ho, Kate and Lee, Robin S. “Insurer Competition in Health Care Markets”. *Econometrica*, 2017, 85, 2, pp. 379–417.
- Ho, Kate and Lee, Robin S. “Equilibrium Provider Networks: Bargaining and Exclusion in Health Care Markets”. *American Economic Review*, 2019, 109, 2, pp. 473–522.
- Hoberg, Gerard and Phillips, Gordon M. (Nov. 2022). *Scope, Scale and Concentration: The 21st Century Firm*. Working Paper.
- Holmes. “The Diffusion of Wal-Mart and Economies of Density”. *Econometrica*, 2011, 79, 1, pp. 253–302.
- Hotz, V. Joseph and Miller, Robert A. “Conditional Choice Probabilities and the Estimation of Dynamic Models”. *The Review of Economic Studies*, 1993, 60, 3, p. 497.
- Houde, Jean-François, Newberry, Peter, and Seim, Katja. “Nexus Tax Laws and Economies of Density in E-Commerce: A Study of Amazon’s Fulfillment Center Network”. *Econometrica*, 2023, 91, 1, pp. 147–190.
- Hsieh, Chang-Tai and Klenow, Peter J. “Misallocation and Manufacturing TFP in China and India”. *The Quarterly Journal of Economics*, 2009, 124, 4, pp. 1403–1448.
- Ifrach, Bar and Weintraub, Gabriel Y. “A framework for dynamic oligopoly in concentrated industries”. *The Review of Economic Studies*, 2017, 84, 3, pp. 1106–1150.
- Illanes, Gaston et al. “A Large-Scale Evaluation of Merger Simulations”, 2024.

- Ito, Koichiro and Zhang, Shuang. "Willingness to Pay for Clean Air: Evidence from Air Purifier Markets in China". *Journal of Political Economy*, 2020, 128, 5, pp. 1627–1672.
- Jeon, Jihye. "Learning and investment under demand uncertainty in container shipping". *The RAND Journal of Economics*, 2022, 53, 1, pp. 226–259.
- Lee, Robin S. "Vertical Integration and Exclusivity in Platform and Two-Sided Markets". *American Economic Review*, 2013, 103, 7, pp. 2960–3000.
- Levinsohn, James and Petrin, Amil. "Estimating Production Functions Using Inputs to Control for Unobservables". *The Review of Economic Studies*, 2003, 70, 2, pp. 317–341.
- Loecker, De. "Product Differentiation, Multiproduct Firms, and Estimating the Impact of Trade Liberalization on Productivity". *Econometrica*, 2011, 79, 5, pp. 1407–1451.
- Loecker, Jan De and Warzynski, Frederic. "Markups and Firm-Level Export Status". *American Economic Review*, 2012, 102, 6, pp. 2437–2471.
- Magnac, Thierry and Thesmar, David. "Identifying Dynamic Discrete Decision Processes". *Econometrica*, 2002, 70, 2, pp. 801–816.
- Miller, Nathan. "Industrial Organization and The Rise of Market Power", 2024.
- Miller, Nathan H. and Weinberg, Matthew C. "Understanding the Price Effects of the MillerCoors Joint Venture". *Econometrica*, 2017, 85, 6, pp. 1763–1791.
- Miravete, Eugenio J., Seim, Katja, and Thurk, Jeff (2026). "Elasticity and Curvature of Discrete Choice Demand Models".
- Murry, Charles. "Advertising in Vertical Relationships: An Equilibrium Model of the Automobile Industry", 2017.
- Nevo, Aviv. "A practitioner's guide to estimation of random-coefficients logit models of demand". *Journal of economics & management strategy*, 2000, 9, 4, pp. 513–548.
- Nevo, Aviv. "Mergers with Differentiated Products: The Case of the Ready-to-Eat Cereal Industry". *The RAND Journal of Economics*, 2000, 31, 3, pp. 395–421.
- Nevo, Aviv. "Measuring Market Power in the Ready-to-Eat Cereal Industry". *Econometrica*, 2001, 69, 2, pp. 307–342.
- Olley, G. Steven and Pakes, Ariel. "The Dynamics of Productivity in the Telecommunications Equipment Industry". *Econometrica*, 1996, 64, 6, pp. 1263–1297.
- Pakes, Ariel, Ostrovsky, Michael, and Berry, Steven. "Simple estimators for the parameters of discrete dynamic games (with entry/exit examples)". *The RAND Journal of Economics*, 2007, 38, 2, pp. 373–399.
- Pavcnik (2003). *Trade liberalization, exit, and productivity improvements: Evidence from Chilean Plants - ProQuest*. en.
- Peter Arcidiacono and Miller, Robert. "Conditional Choice Probability Estimation of Dynamic Discrete Choice Models With Unobserved Heterogeneity". *Econometrica*, 2011, 79, 6, pp. 1823–1867.
- Petrin, Amil. "Quantifying the Benefits of New Products: The Case of the Minivan". *Journal of Political Economy*, 2002, 110, 4, pp. 705–729.
- Pinkse, Joris, Slade, Margaret E., and Brett, Craig. "Spatial Price Competition: A Semiparametric Approach". *Econometrica*, 2002, 70, 3, pp. 1111–1153.
- Rubens, Michael. "Market Structure, Oligopsony Power, and Productivity". *American Economic Review*, 2023, 113, 9, pp. 2382–2410.
- Rubens, Michael. "Labor-Market Power, Deadweight Loss, and Technology Adoption", 2025.
- Rust, John. "Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher". *Econometrica*, 1987, 55, 5, p. 999.
- Ryan, Stephen P. "The Costs of Environmental Regulation in a Concentrated Industry". *Econometrica*, 2012, 80, 3, pp. 1019–1061.
- Sabal, Alejandro (Jan. 2025). "Product Entry in the Global Automobile Industry".
- Seim, Katja. "An empirical model of firm entry with endogenous product-type choices". *The RAND Journal of Economics*, 2006, 37, 3, pp. 619–640.

- Shapiro, Carl. “Protecting Competition in the American Economy: Merger Control, Tech Titans, Labor Markets”. *Journal of Economic Perspectives*, 2019, 33, 3, pp. 69–93.
- Singleton, John D. “Incentives and the Supply of Effective Charter Schools”. *American Economic Review*, 2019, 109, 7, pp. 2568–2612.
- Sweeting, Andrew. “Dynamic Product Positioning in Differentiated Product Markets: The Effect of Fees for Musical Performance Rights on the Commercial Radio Industry”. *Econometrica*, 2013, 81, 5, pp. 1763–1803.
- Syverson, Chad. “Macroeconomics and Market Power: Context, Implications, and Open Questions”. *Journal of Economic Perspectives*, 2019, 33, 3, pp. 23–43.
- Wollmann, Thomas G. “Trucks without Bailouts: Equilibrium Product Characteristics for Commercial Vehicles”. *American Economic Review*, 2018, 108, 6, pp. 1364–1406.
- Xia, Tianli. “Welfare Effects of Resale Price Maintenance: Evidence from the Chinese Pharmaceutical Industry”, 2024.

Preparing for a Meeting with Me (from John Asker)

Meetings with faculty matter, and when they are well run they are among the most useful hours you will spend. Ph.D. students, however, are rarely given any guidance on how to run one. What follows is what I expect, with an emphasis on early-stage meetings in which we are discussing a research idea that is still taking shape.[†]

Know what you want out of the meeting

The first thing I will ask is what you are hoping to get out of our time together. Have an answer ready. A meeting without a purpose tends to drift, and half an hour disappears quickly.

Come with something written

Do not ask for a meeting until you have a short research proposal, roughly four to six pages, that you can send me in advance. If you request a meeting without one, I will most likely ask you to write it first, and we will set a time once it exists.

This is not a hurdle I have invented for students. It is what I do before I take an idea to a colleague, and it is a habit worth acquiring early.

What the proposal should address

You should have thought about each of the following:

1. What, precisely, is the research question?
2. Why is it interesting or important?
3. What does the project contribute to the existing economics literature?
4. What is the current state of that literature?
5. Which industry or empirical setting do you have in mind?
6. How does that setting actually work?
7. Has anyone else studied it, and what did they do?
8. What data are available to you?

[†]The substance of this policy follows advice given by John Asker (UCLA) to his own students; see <http://www.johnasker.com/Students.html>.

9. How do you propose to structure an answer to the question?
10. What sort of model do you have in mind?
11. If the project is empirical: what is the basic approach — reduced form, structural, or some combination — and what are the steps involved?
12. How do the data fit into that approach?
13. What difficulties do you expect to run into?

Your answers do not need to be right, or even complete. They need to be thought through far enough that you can hold up your end of an informative half-hour conversation, and the document should show that the thinking actually happened.

On having bad ideas

Most of my own research ideas are poor ones, and you should not be afraid to have poor ones too. Telling good ideas from bad is a skill, and it takes time to develop: an idea you have written off may turn out to have something in it, and one you are excited about may not survive contact with a second reader. Writing the idea down and then arguing about it with someone else is the fastest way to find out which is which. In my experience roughly half of my ideas die in the writing; of those that survive, roughly half die in the first conversation.